

Flooring Dance: From Perception to Inscription in a Culturally Situated Embodied Geometry Design

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Solving geometry problems often depends on introducing auxiliary lines to reveal latent structural relations among figurally disparate diagrammatic components. Yet generating such lines is challenging, as students must envision new elements that are not yet materially available. Meanwhile, a phenomenologically comparable dynamic occurs in movement learning, such as in dance: to coordinate the enactment of independent multi-limb actions, dancers generate *attentional anchors*—imagined perceptual structures that coherently organize kinesiological intentionality. Building on Reed and Bril's theoretical construct *field of promoted action*, the Geometry Resources in Dance (GRiD) design-based research project conjectures that, given appropriate cultural mediation that includes semiotic resources and attentive guidance, students can objectify their covert attentional anchors from dance practice as overt auxiliary lines for geometry discourse. We report on results from micro-analyzing the multimodal social interactions of four Balinese 5th-grade students as they coordinated their actions to form a circle in a dancing context. Attentional anchors, first evidenced in their gaze and bodily orientation, gradually materialized in response to joint-action constraints into a geometric system of auxiliary lines inscribed on a provided floor-mat template. We speculate that the Balinese cultural ecology—where sitting, kneeling, and working on the ground are normative—facilitated the literal grounding of perception as geometric inscription. Within the GRiD context, action and inscription are not dichotomized but epistemically continuous: tracing lines of perceptual intentionality on the floor is experienced not as mathematizing, modeling, or abstracting but as gestured marking of enactive practice.

Keywords: auxiliary lines, attentional anchors, culturally situated learning, embodied mathematics design

Extended Summary

We address the challenge of teaching and learning auxiliary lines, which stems from students' difficulty envisioning figural components that are not perceptually available (Foster, 2023). Our inquiry turned to movement pedagogy, where dancers rely on *attentional anchors*—perceptual structures that guide coordinated action in complex movements (Hutto & Sánchez-García, 2015). Building on the construct of the *field of promoted action* (Reed & Bril, 1996), we propose that, with cultural mediation, students can transform attentional anchors from dance into auxiliary lines for geometry discourse.

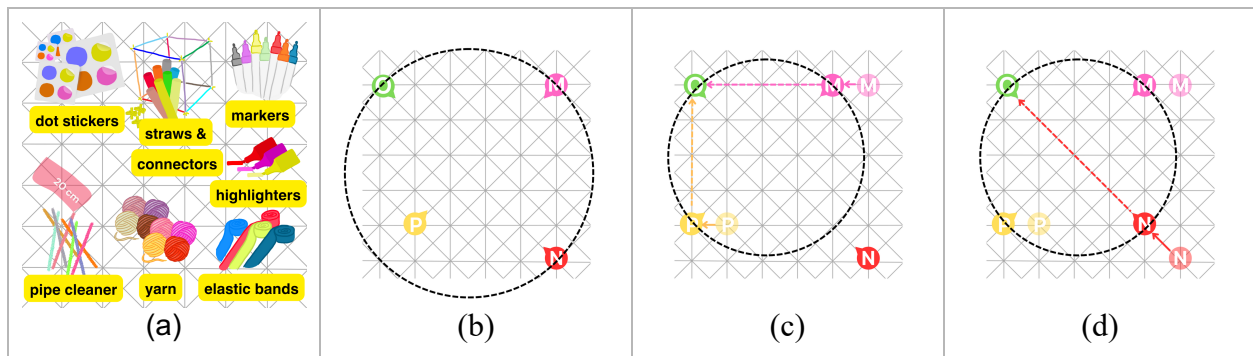


Figure 1. Semiotic resources of the dance–geometry activities. (a) GRiD diagrammatic mat and marking accessories. (b–d) Schematic of four students’ positions while forming a circle: initialed dots with triangles (“noses”) show torso orientations; faded dots mark prior positions; solid arrows indicate displacements; dashed arrows indicate movement intentionality.

We developed an action-based embodied mathematics design (Abrahamson, 2014), Geometry Resources in Dance (GRiD), which integrates movement activities with diagrammatic resources (Figure 1a) rooted in Balinese traditions (Apsari & Abrahamson, 2024, 2025). In Balinese dance training, teachers refine students’ movements through narrative guidance, situating performance within Bali’s socio-spiritual contexts. Such pedagogical cueing functions as attentional anchors, facilitating coordinated movement. For instance, instructors reference the traditional decorative stage (*kalangan*) to orient dancers’ posture and direction (Bandem & deBoer, 1978).

We conducted a recorded teaching experiment with 76 fifth-grade students (ages 9–12) in Bali, Indonesia. This paper focuses on one group of four—Merry, Nath, Owen, and Philip—who silently micro-adjusted their positions on the mat (Figure 1b–d), creating a circle formation. Although familiar with the shape, students had not formally studied the circle’s elements (e.g., center, radius, diameter) or definition. Microgenetic (Siegler, 2006) and multimodal conversation analysis (Mondada, 2019) of students’ embodied, discursive, and inscriptive work traces how attentional anchors emerged, were negotiated, and materialized as shared auxiliary lines grounding the discovery of geometrical concepts.

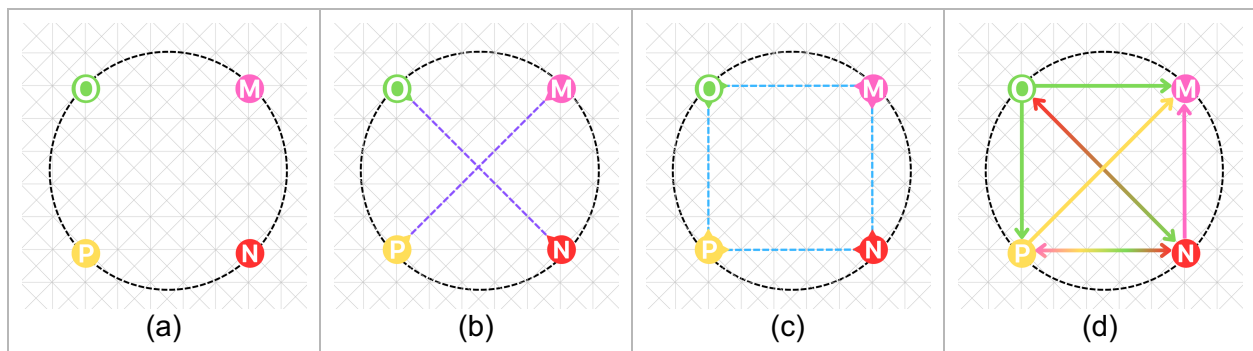


Figure 2. Summary of students’ co-constructed auxiliary lines for evaluating a circle formation on the GRiD mat. (a) Students’ positions on the constructed circle. (b) *Opposites* strategy. (c) *Adjacents* strategy. (d) Final collective auxiliary lines.

Having formed a circle, participants sat and marked their positions on the GRiD mat (Figure 2a). Their discussion revealed that they had previously operated in different orientations (*heteroperception*, Abrahamson et al., 2025): some oriented to peers opposite them, while others aligned with adjacent peers (Figure 2b–c). When prompted to explain, students’ implicit heteroperception surfaced explicitly through gestures, verbalizations, and eventually was inscribed as a coherent system of pooled (Rączaszek-Leonardi et al., 2014) auxiliary lines on the GRiD mat (Figure 2d).

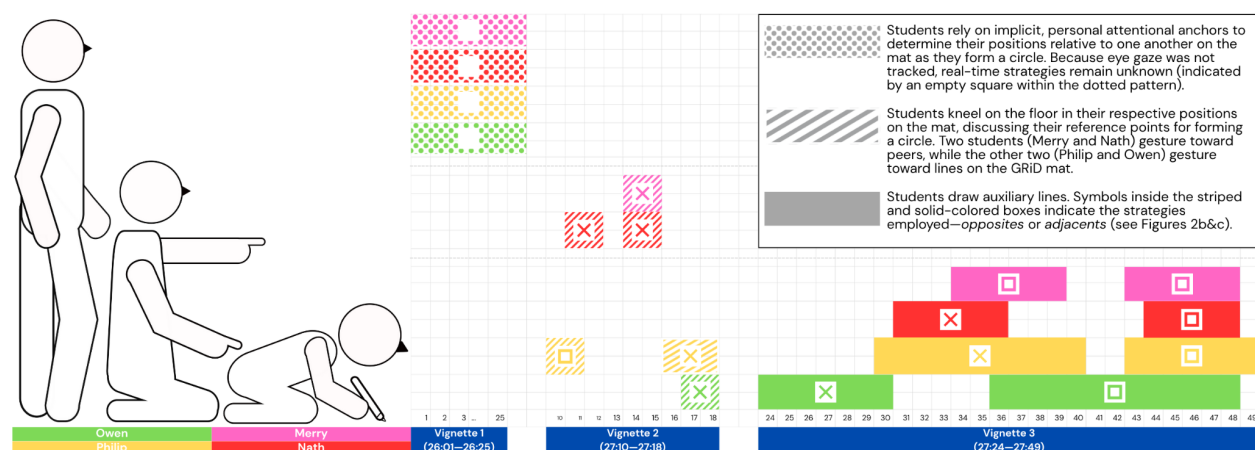


Figure 3. Seeing, pointing, tracing. Temporal progression of attentional anchors from perceptual intentionality of action (enacting a circle) to materialized auxiliary lines (evaluating it).

The diagrammatic mat enabled students to propose, examine, and adapt one another’s contributions, supporting the evolution of attentional anchors into auxiliary lines (Figure 3). Ephemeral gestures shifted into durable inscriptions (Vignette 2), and the initially overlooked *adjacents* strategy (Vignette 2) resurfaced when Merry introduced the line between Nath and herself, prompting others to follow (Vignette 3). The continuity from enactment to inscription was fostered by Balinese cultural ecology, where dance emphasizes grounding in the earth and floor-sitting is a social norm, reflecting a close relation with nature (Dibia & Ballinger, 2023). In GRiD, students performed the dance formation and immediately marked their positions on the same mat—tracing lines on the floor was experienced not as abstraction but as gestured marking of enactive practice (Noss & Hoyles, 1996).

Our findings support legitimizing cultural norms as generative resources in mathematics education. With semiotic resources such as the GRiD mat, students enacted, evaluated, and negotiated strategies that transformed attentional anchors into auxiliary lines while remaining in their circle positions (see Abrahamson et al., 2011). By contrast, relocating to a desk or board would have disrupted their embodied relation to the circle. We hope this study stimulates dialogue on mathematical epistemology and the role of media in sustaining phenomenological continuity from action to symbol (see Bamberger, 2020).

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